UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
2022/2023 SECOND SEMESTER EXAMINATION

COURSE:	MTH222: REAL ANALYSIS I
DATE SCHEDULED:	MONDAY 15th JULY, 2024
TIME ALLOWED:	TWO HOURS (2 HRS)

Instructions: Attempt any four (4) questions only.

1. Define the following terms:
   (a) (i) Monotone sequence (ii) Monotone non-decreasing sequence (iii) Monotone non-increasing sequence
   (b) Verify if the sequences defined by (i) $\{a_n\} = \{1 - \frac{1}{n}\}$ (ii) $\{b_n\} = \{(-1)^n\}$ are monotone non-decreasing, monotone non-increasing or not monotone sequences.
   (c) Prove that the $\lim_{x \to 3} (x^2 + 2x) = 15$.

2. (a) Prove that if a sequence converges to a limit, then that limit is unique.
   (b) (i) Show that the sequence $\{\frac{n-7}{2-3n}\}_{n=1}^{\infty}$ converges.
   (ii) Prove that the $\lim_{n \to \infty} \sqrt{\frac{n^2}{n^2+1}} = 1$.

3. (a) (i) Find the limit of $\frac{1+\sin n}{n}$ as $n \to \infty$. (ii) Find the limit of $\frac{2+\cos n}{n+1}$ as $n \to \infty$.
   (b) (i) When is a sequence said to be bounded.
   (ii) For each of the set below, say if it is bounded below, bounded above, not bounded below, not bounded above. Find the infimum and supremum of each. (a) $[0, 1]$ (b) $\{\frac{1}{n} : n \in \mathbb{N}\}$ (c) $(-\infty, \infty)$.

4. (a) Explicitly define the following:
   (i) Nested interval.
   (ii) Every convergence sequence is a Cauchy sequence, prove.
   (b) (i) Prove that the sequence $\{a_n\} = \{\frac{1}{n}\}_{n=1}^{\infty}$ is Cauchy sentence and deduce that it does not converge in $(0, 1)$.
   (ii) Evaluate (a) $\lim_{x \to 0} \frac{e^{2x}-1}{x}$ (b) (a) $\lim_{x \to 0} \frac{\sin x - x\cos x}{x^2 \sin x}$.

5. (a) State and verify the MVT for $f(x) = \sqrt{x-1}$ on $[1, 3]$.
   (b) State Rolle's Theorem and verify Rolle's Theorem for $f(x) = x^2(1-x)^2$, $0 \leq x \leq 1$.
   (c) Show that the result of Rolle's theorem is not true for $f(x) = 2x^{-2}$. on $[-1, 1]$.

6. (a) Define the following: (i) infimum of set, (ii) supremum of set.
   (b) Prove that the sup $\{\frac{n}{2n+1}, n = 1, 2, \ldots\} = \frac{1}{2}$.
   (c) If a and b are any two real numbers, prove that (i) $|a - b| \geq ||a| - |b||$.